



1. Identification of the substance/preparation and of the company undertaking

Trade Name: KPL Zinc Phosphate

Application: Used as a corrosion-inhibitive pigment in primers, industrial coatings, and water-based paints

Manufacturer/Supplier:

Kesar Petroproducts Ltd.

Plot No. D-7/1, Lote Parshuram MIDC, Tal-Khed, Dist. Ratnagiri, Maharashtra – 415722

Further information: Product Safety Department, Kesar Petroproducts Ltd., Ratnagiri Plant

2. Hazards Identification

Classification: Not classified as hazardous under GHS (Globally Harmonized System) as per current knowledge

Hazard Statements: Not applicable

Precautionary Statements: Avoid dust formation; use general industrial hygiene practices

3. Composition/Information on Ingredients

Description: Zinc Phosphate

Chemical Formula: $\text{Zn}_3(\text{PO}_4)_2 \cdot 2\text{H}_2\text{O}$

CAS No.: 7779-90-0

Purity: ≥98% (typical industrial grade)

4. First-Aid Measures

Inhalation: Move to fresh air. If breathing is difficult, give oxygen. Seek medical attention if symptoms persist.

Skin Contact: Wash thoroughly with soap and water. Remove contaminated clothing.

Eye Contact: Rinse cautiously with plenty of water for at least 15 minutes. Remove contact lenses if present. Seek medical advice.

Ingestion: Rinse mouth. Do NOT induce vomiting. Seek immediate medical attention.

5. Fire-Fighting Measures

Extinguishing Media: Water spray, CO₂, dry chemical, or foam

Special Hazards: In fire, may release zinc oxide and phosphorus oxides (toxic fumes)

Special Procedures: Wear self-contained breathing apparatus and full protective clothing

6. Accidental Release Measures

Personal Precautions: Wear protective gloves, safety goggles, and dust mask

Environmental Precautions: Prevent entry into drains, sewers, or waterways

Cleanup Methods: Collect mechanically or by wet method; avoid dry sweeping. Place in sealed container for disposal

7. Handling and Storage

Handling: Avoid dust generation. Use in well-ventilated area. Ground containers during transfer.

Storage: Store in cool, dry, well-ventilated area. Keep in original, tightly closed containers. Protect from moisture.

8. Exposure Controls/Personal Protection

Occupational Exposure Limit:

Zinc (as Zn): TLV-TWA 5 mg/m³ (ACGIH)

Phosphates: No specific limit; general particulate limit applies

Engineering Controls: Local exhaust ventilation recommended

Personal Protective Equipment:

Respiratory: NIOSH-approved dust mask (N95 or equivalent) if airborne dust > permissible levels

Hand Protection: Nitrile or rubber gloves

Eye Protection: Safety goggles

Skin Protection: Protective clothing to prevent dust contact

9. Physical and Chemical Properties

Form: Fine white powder

Odour: Odourless

pH (10% aqueous suspension): 5.5 – 7.0

Specific Gravity: ~3.5

Solubility in Water: Slightly soluble (<1 g/L)

Melting Point: Decomposes before melting (~900°C)

Flash Point: Not applicable

Explosive Properties: Dust may form explosive mixture in air

10. Stability and Reactivity

Stable under normal conditions

Incompatible with: Strong acids (may release toxic phosphine gas), strong oxidizing agents

Hazardous Decomposition Products: Zinc oxide fumes, phosphorus oxides (under fire conditions)

11. Toxicological Information

Acute Toxicity (oral, rat): LD₅₀ > 2,000 mg/kg (low toxicity)

Skin Irritation: Not classified as irritant

Eye Irritation: May cause mechanical irritation due to particulates

Sensitization: Not known to be a skin sensitizer

Chronic Effects: Prolonged inhalation of dust may cause respiratory irritation

12. Ecological Information

Toxicity to Aquatic Life: Zinc compounds are toxic to aquatic organisms

Persistence: Zinc ions may accumulate in sediments

Precaution: Do not discharge into environment

13. Disposal Considerations

Product: Dispose of as non-hazardous industrial waste only after confirming local regulations

Contaminated Packaging: Rinse thoroughly before recycling or disposal

Note: Zinc content may classify as hazardous waste in some jurisdictions—consult local authority

14. Transportation Information

UN Number: Not regulated

Proper Shipping Name: Zinc phosphate

Transport Classification: Not restricted by IMDG, IATA, ADR, or RID

15. Regulatory Information

GHS Classification: Not classified as hazardous under EU CLP or OSHA HCS

REACH: Registered (if applicable; confirm based on annual tonnage)

TSCA: Listed

16. Other Information

This information is based on current scientific knowledge and is believed to be accurate. However, it does not constitute a warranty or establish a legally binding agreement. Users must ensure compliance with local laws.

Issued by: Quality Control Department, Kesar Petroproducts Ltd.

Date: 29 December 2025